

Local AI, Models & Model Routing

Research objective

- Understand how multiple local AI capabilities could theoretically support the workbench and what the actual constraints are.
- The objective is to understand **why a particular capability may require a particular model class**.

Research areas

Local model execution

- Local LLM inference
- Open-weight models
- On-premise inference
- Air-gapped inference
- Local VLM inference
- Quantized inference
- GPU inference
- CPU/GPU hybrid inference
- VRAM requirements
- Context-length constraints
- Inference latency
- Throughput

Model capabilities

Research models according to **capability**, not popularity:

- General reasoning
- Coding
- Document understanding
- Vision
- OCR
- Multimodal reasoning
- Structured output
- Tool calling

Multi-model system

- Model specialization
- Model capability profiles

- Multiple local models
- Model lifecycle
- Model loading/unloading
- Model switching
- Model fallback
- Model resource management

Automatic routing

Research:

- Task classification
- Capability classification
- Model routing
- Model selection
- Model cascades
- Routing based on quality
- Routing based on latency
- Routing based on available GPU resources
- Routing failure
- Incorrect model selection
- Model-routing evaluation

Keywords

R4 — Technical Research

Local Model Execution

- local LLM inference architecture
- on-premise LLM inference
- air-gapped LLM inference
- offline LLM inference
- open-weight model deployment
- local vision-language model deployment
- local multimodal model inference
- GPU-based local LLM inference
- CPU-GPU hybrid LLM inference
- quantized LLM inference
- low-VRAM LLM inference
- mid-range GPU LLM deployment

Model Serving

- local LLM serving architecture

- multi-model serving architecture
- model lifecycle management
- local model registry
- dynamic model loading
- model hot swapping
- concurrent local model serving
- LLM inference resource scheduling
- GPU memory management for LLM serving

Model Routing

- LLM model routing architecture
- automatic model selection
- task-based model routing
- capability-based model routing
- semantic task classification for LLM routing
- multi-model agent architecture
- heterogeneous LLM orchestration
- model routing evaluation
- model selection accuracy
- model routing latency
- model capability profiling
- local model fallback strategies
- model routing under GPU constraints

Feasibility

- LLM VRAM requirements
- VLM VRAM requirements
- quantization impact on reasoning
- quantization impact on vision
- inference latency benchmarking
- local model throughput
- context length GPU memory
- KV cache memory optimization
- model performance on mid-range GPUs

For every candidate model, record

Model

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Primary capability

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Input types



Output types



Hardware requirement



VRAM requirement



Inference characteristics



Known limitations



Suitability for project